

Approach-Region Commitment and Environmental Friction: A Cross-Phase Damping Atlas for Thermally Activated Chemical Reactions

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Abstract

The commitment of a reacting system to productive barrier crossing depends on how strongly the environment dissipates energy into the approach-region coordinate, rather than oscillatory recrossing. This dissipation is quantified by a single dimensionless damping ratio, $\chi_0 = \Gamma_{\text{eff}}/(2\Omega_0)$, where Ω_0 is the precursor-well approach frequency and Γ_{eff} is the total projected friction. Using three independent estimator routes calibrated to spectroscopic linewidths (Route A), solvent longitudinal relaxation times (Route B), and DFT electronic friction coefficients (Route C), χ_0 is computed here for nine canonical systems, constituting an Approach-Region Commitment (ARC) Atlas, spanning gas-phase unimolecular reactions, liquid-phase electron transfer and proton-coupled electron transfer, surface heterogeneous catalysis, and solid-state ion transport. Gas-phase systems are deeply underdamped ($\chi_0 \sim 10^{-3}$ – 10^{-2}), providing a dynamical interpretation of the excess-energy requirements of RRKM unimolecular kinetics through the approach-region commitment probability rather than through transition-state recrossing alone. Selected condensed-phase systems approach or enter the near-critical window $\chi_0 \in [0.8, 1.3]$ (K-promoted metal surfaces; fast-solvent ET near the upper window flank), while viscous solvents and poisoned surfaces fall into the overdamped regime. The results are not obtained by fitting χ_0 to reaction outcomes: Γ_{eff} and Ω_0 are taken from independently measured, computed, or explicitly bracketed physical observables in each case. The progression from gas to liquid to surface corresponds to a several-order-of-magnitude span in χ_0 , revealing that solvent choice, surface promotion, and phase selection each shift χ_0 in a predictable direction relative to the exceptional-point stability boundary at $\chi_0 = 1$.

Keywords: approach-region dynamics; damping ratio; barrier crossing; cross-phase chemistry; electron transfer; heterogeneous catalysis; RRKM theory; PCET

Introduction

Rate theory for thermally activated barrier crossing has two well-developed pillars. The thermodynamic pillar encompasses transition-state theory (TST), Marcus reorganization energies, Brønsted–Evans–Polanyi (BEP) scaling relations, and Sabatier-type volcano plots, describes the probability that a reactive system reaches the transition state at all. The dynamical pillar encompasses Kramers friction theory, Grote–Hynes non-Markovian corrections, Zusman solvent-friction theory, and Newns–Anderson electronic friction, describes what happens to the reactive coordinate once it enters the barrier region.

What has been missing is a single, measurable descriptor whose mathematical form is invariant across phases under linearization of the precursor-well generator, quantifying how efficiently the *approach* to the barrier converts available thermal energy into irreversible commitment to products. Existing descriptors are each domain-specific: τ_L is used in electron transfer but not in surface catalysis; the Newns–Anderson width Γ_{NN} is used in surface chemistry but not in solution; RRKM density of states is used in gas-phase

kinetics but has no direct analogue in condensed phase.

Three independently derived theoretical frameworks, Pollak–Grabert–Hänggi (PGH) barrier-crossing theory, Zusman electron-transfer theory, and Newns–Anderson surface-chemistry theory, each contain a locally equivalent damped-oscillator generator in their approach-region dynamics. When expressed in terms of the dimensionless damping ratio $\chi_0 = \Gamma_{\text{eff}}/(2\Omega_0)$, the same exceptional-point boundary at $\chi_0 = 1$ appears in the linearized commitment problem across all three domains. This structural invariance reflects the shared form of the linearized generator; the physical origin of Γ_{eff} differs completely across phases, but the dimensionless ratio captures the dynamical regime in each case. Three calibrated estimator routes connect χ_0 to spectroscopic linewidths (Route A), solvent relaxation times (Route B), and DFT electronic friction coefficients (Route C).

The present paper asks whether the same descriptor organizes barrier-crossing dynamics across the chemical phases examined here: gas-phase unimolecular reactions without any environmental friction; liquid-phase electron transfer and proton-coupled electron transfer (PCET); and solid-state systems including a heterogeneous catalyst and an ionic conductor. The answer is yes, but the structure of the result is richer than a simple clustering of all systems near $\chi_0 \approx 1$. The atlas is descriptive rather than normative: it does not imply that all reactions must satisfy $\chi_0 \approx 1$ for efficient chemistry, only that χ_0 organizes the approach-region dynamical regime across the systems studied.

Gas-phase systems are deeply underdamped ($\chi_0 \sim 10^{-3}$), not because the approach-region framework fails but because isolated molecules have access only to intramolecular vibrational redistribution (IVR) as a dissipation channel, and IVR provides orders of magnitude less friction than a solvent bath or a metal surface. This underdamped character is quantitatively consistent with the classical RRKM finding that unimolecular reactions require substantial excess energy above threshold to achieve near-unit commitment probability: an underdamped approach coordinate recrosses repeatedly before committing, requiring the system to sample the transition-state region many times.

Condensed-phase systems span the full range from underdamped (CO on Cu) through near-critical (N₂ on K-promoted Fe) to overdamped (ET in viscous solvents), with known high-performance cases moving toward the near-critical window while underdamped and overdamped limits correspond to recognized dynamical failure modes. The progression from gas to liquid to surface is therefore a χ_0 progression, a systematic increase in environmental friction from IVR alone ($\chi_0 \sim 10^{-3}$) through solvent-mediated friction ($\chi_0 \sim 1$) toward electronically dominated friction in the strongest surface-adsorbate systems.

Beyond organizing existing data, the ARC Atlas provides an immediately applicable calculation pathway: three estimator routes that locate any new thermally activated system on the χ_0 axis using independently measurable observables, without free parameters or system-specific fitting.

The paper is organized as follows. Section 1 summarizes the framework, estimator routes, and evidence grading. Sections 2, 3, and 4 present the solid-state/surface, liquid-phase, and gas-phase atlas entries, respectively. Section 5 synthesizes the cross-phase picture and identifies prospective tests.

The nine systems were chosen according to three explicit criteria: (i) each is a canonical archetype in its subfield rather than an obscure edge case; (ii) each has independently measured, computed, or explicitly bracketed inputs for Γ_{eff} and Ω_0 , with all non-prospective entries satisfying at least Tier 2 evidence for regime classification; and (iii) each represents a phase or mechanism not covered by the other eight entries. Li⁺/LLZO is retained as an explicitly marked Tier 3 prospective stress test, not as a validated result. Five of the nine systems are underdamped, three are overdamped, and only one (N₂/K-Fe(111)) is inside the

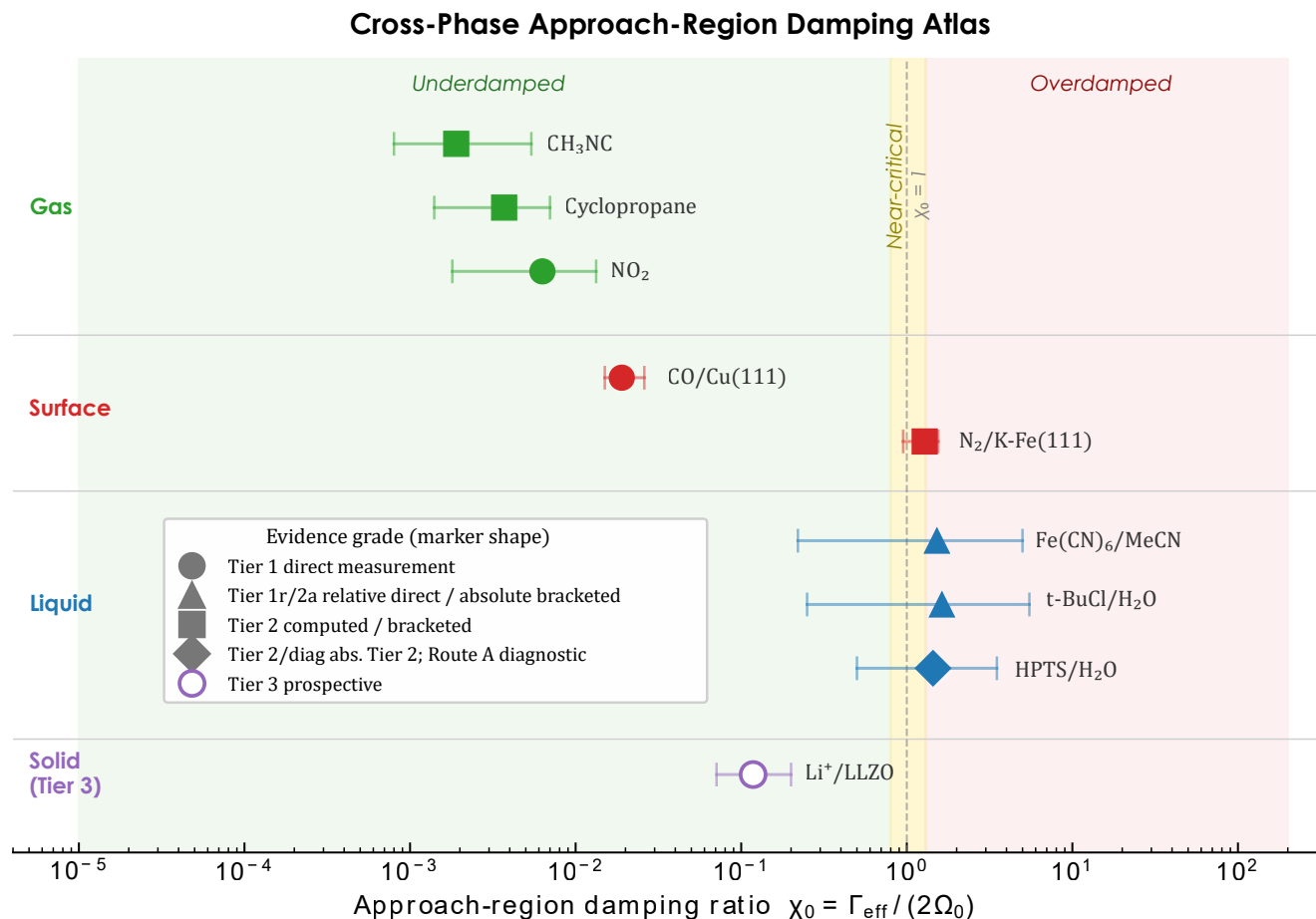


Figure 1: Cross-phase approach-region damping atlas. All nine systems are placed on a single log-scale χ_0 axis grouped by phase. Marker shape encodes evidence grade: filled circle (Tier 1, direct measurement), filled triangle (Tier 1r/2a, relative direct / absolute bracketed), filled square (Tier 2, computed or bracketed), filled diamond (Tier 2/diag, absolute Tier 2; Route A diagnostic), open circle (Tier 3, prospective). Marker color encodes phase: green (gas), red (surface), blue (liquid), purple (solid). Horizontal error bars show the full uncertainty bracket for each system. Shaded regions: underdamped ($\chi_0 < 0.8$, green), near-critical ($0.8 \leq \chi_0 \leq 1.3$, gold), overdamped ($\chi_0 > 1.3$, red). No χ_0 value is fitted to a reaction outcome; all inputs are independently measured, computed, or explicitly bracketed quantities. Only N₂/K-Fe(111) is inside the near-critical window; the gas-phase systems are underdamped by two to three orders of magnitude, and the fastest liquid-phase solvents are slightly overdamped.

near-critical window $[0.8, 1.3]$. Systems considered but excluded include enzyme-catalyzed barrier crossing, where multidimensional coordinate coupling makes single- Ω_0 extraction model-dependent, and excited-state photoisomerizations, where the approach-well topology differs from ground-state thermal activation. The set is intended to demonstrate cross-phase breadth.

Figure 1 summarizes the complete nine-system atlas that results from the analysis below: all systems on a shared log-scale χ_0 axis, with marker shape encoding evidence grade and color encoding phase. The five-order-of-magnitude span motivates the framework and system-by-system analysis that follow.

1 Theoretical Framework

The approach-region damping ratio

Full derivation is given in the companion paper [27]. The quantities required for the atlas are summarized here.

The central quantity is the dimensionless approach-region damping ratio:

$$\chi_0 = \frac{\Gamma_{\text{eff}}}{2\Omega_0}, \quad (1)$$

where Ω_0 ($\Omega_0^2 > 0$) is the precursor-well approach frequency and Γ_{eff} is the total projected dissipation coupling the approach coordinate to its bath. The exceptional-point boundary at $\chi_0 = 1$ marks the transition between oscillatory underdamped approach dynamics ($\chi_0 < 1$, recrossing-prone) and monotonic overdamped dynamics ($\chi_0 > 1$, sluggish commitment). The near-critical window $\chi_0 \in [0.8, 1.3]$ is defined operationally by the criterion $\eta(\chi_0) \geq 0.95$, where $\eta(\chi) = 2\chi/(1 + \chi^2)$ is the reduced dynamical efficiency (Figure 2), derived as the first-passage efficiency of the linearized precursor-well generator: the fraction of approach-coordinate trajectories that commit to product rather than recross, evaluated at the exceptional-point boundary [27].

The exceptional-point boundary at $\chi_0 = 1$ arises from the linearized precursor-well generator $\ddot{q} + \Gamma_{\text{eff}}\dot{q} + \Omega_0^2 q = 0$: in plain terms, $\chi_0 = 1$ is the friction level that maximizes commitment probability by eliminating oscillatory recrossing without entering the sluggish overdamped regime. At $\chi_0 < 1$ the eigenvalues are complex conjugates (oscillatory approach, recrossing-prone); at $\chi_0 = 1$ the two eigenvalues coalesce (exceptional point, fastest non-oscillatory approach); at $\chi_0 > 1$ both eigenvalues are real and negative (overdamped, sluggish commitment). The maximal dynamical efficiency $\eta = 1$ at $\chi_0 = 1$ follows directly from this coalescence. The exceptional-point interpretation across broader physical systems spanning quantum dissipation to cosmological acceleration is developed in the companion paper [26].

Two linearization caveats apply. First, the exceptional-point analysis is exact for the linear generator and identifies an operational boundary for anharmonic systems where the precursor well is approximately harmonic throughout the approach region; strongly anharmonic wells shift the true dynamical optimum. Second, this framework uses the approach-well frequency Ω_0 , while Kramers escape-rate theory uses the barrier-top frequency ω_b [21]. For systems where the precursor well is approximately harmonic throughout the approach ($\Omega_0 \approx \omega_b$), the two optimal-friction conditions converge; for strongly anharmonic wells the conditions diverge and the near-critical window $[0.8, 1.3]$ should be understood as an operational guideline for the linearized case.

Relationship to variational transition-state theory (VTST). In VTST, a transmission coefficient $\kappa \leq 1$ corrects the TST rate for dynamical recrossing [20]. κ is a trajectory-integrated correction applied post-hoc to a known barrier: it requires computing the full reaction-coordinate dynamics and is not directly portable across phases. χ_0 is complementary rather than redundant: it is computed solely from approach-region observables (Γ_{eff} and Ω_0) before knowing the rate outcome, it is cross-phase by construction, and it predicts a dynamical regime rather than correcting a rate. In the Kramers turnover regime, κ and χ_0 converge numerically [21]; χ_0 extends the concept to surface electronic friction and gas-phase IVR where trajectory-based κ is computationally inaccessible or physically ill-defined.

Note on Route A and CO/Cu(111): Route A measures the Lorentzian linewidth of the adsorbed Cu-

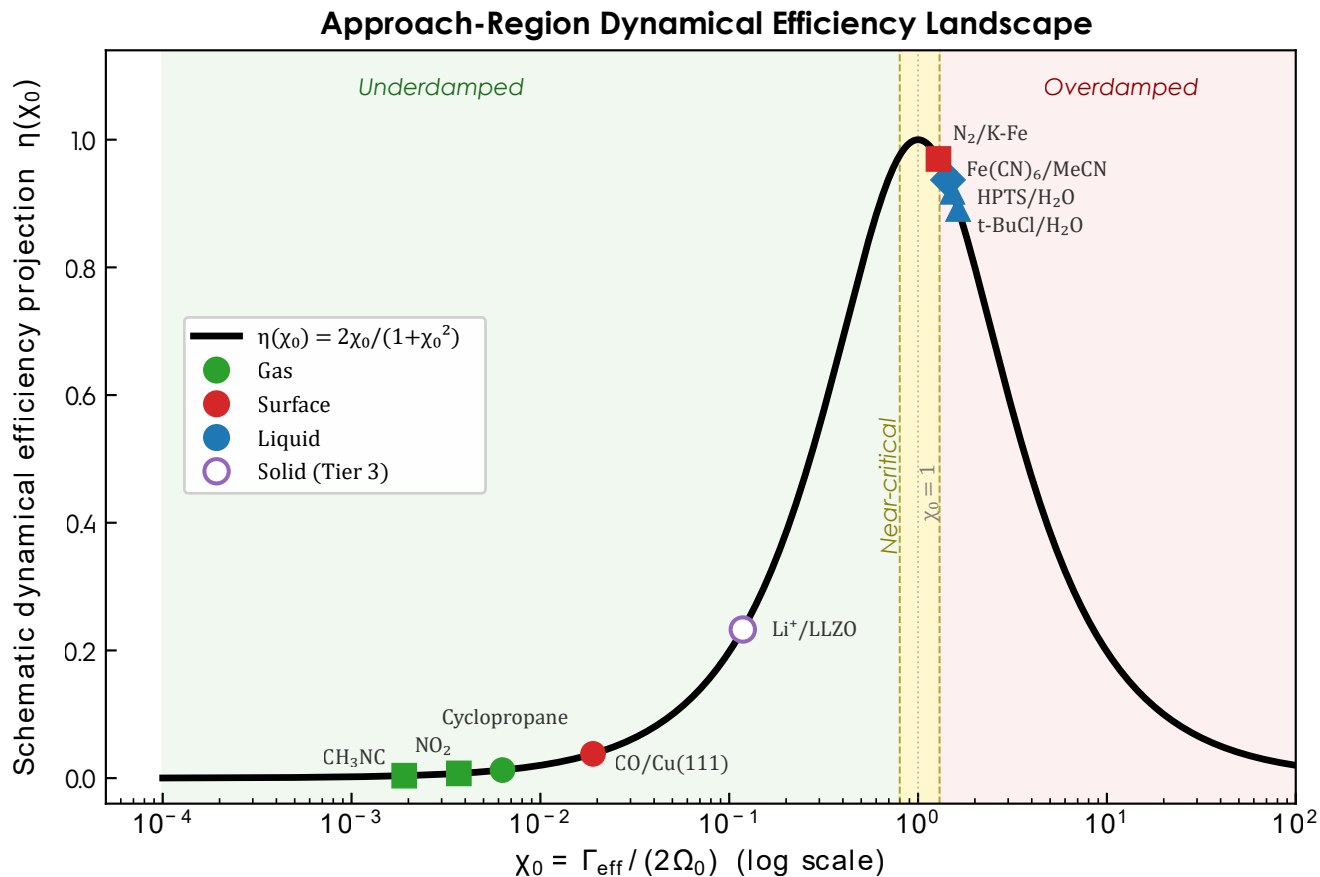


Figure 2: Approach-region dynamical efficiency landscape. The curve $\eta(\chi_0) = 2\chi_0/(1 + \chi_0^2)$ is plotted on a log scale with all nine atlas systems overlaid at their central χ_0 values. The y-axis is a schematic dynamical efficiency projection, not normalized experimental performance. Marker shape encodes evidence grade; marker color encodes phase (green: gas, red: surface, blue: liquid, purple: solid; open circle: Tier 3 prospective). The near-critical window $[0.8, 1.3]$ (gold band, $\eta \geq 0.95$ throughout) encompasses the exceptional-point maximum at $\chi_0 = 1$. Gas-phase systems occupy the deeply underdamped left flank; viscous-solvent systems fall to the overdamped right. See Section 2–4 for system details and Table 1 for all values.

CO perpendicular stretch in the physisorption well. This characterizes the electronic friction experienced by the adsorbate-surface bond coordinate during the final commitment step from physisorbed precursor to chemisorbed product. It does not measure friction during gas-phase approach to the physisorption well; that approach involves corrugation and phonon excitation. The precursor-well picture identifies the physisorption minimum as the approach-region generator for the commitment step, making the adsorbed-state linewidth the appropriate friction input for Route A in surface systems.

χ_0 provides a strictly reaction-dynamical coordinate that contributes to the pre-exponential commitment probability among systems with comparable thermodynamic descriptors, complementing Marcus reorganization energies, BEP scaling relations, d-band metrics, and volcano plots rather than displacing them.

Estimator routes

Route A: spectroscopic.

$$\chi_{\text{spec}} = \frac{\Delta\tilde{\nu}_L}{2\tilde{\nu}}, \quad (2)$$

where $\Delta\tilde{\nu}_L$ is the Lorentzian component of the IR or Raman linewidth (extracted from a Voigt fit) and

$\tilde{\nu}$ is the mode frequency. When the lineshape exhibits Fano asymmetry (interference with a continuous electronic or phonon background), the intrinsic homogeneous width Γ is extracted from a direct Fano fit: $I(E) \propto (q + \epsilon)^2 / (1 + \epsilon^2)$, $\epsilon = 2(E - E_R) / \Gamma$. The asymmetry parameter q and the linewidth Γ are independent quantities: on metallic surfaces Fano profiles arise from coupling to the electron-hole pair continuum [17, 18], which contributes to Γ , but the *magnitude* of Γ is not determined by q alone. Fano asymmetry indicates the presence of non-adiabatic electronic coupling; it does not by itself establish that electronic friction dominates over phononic contributions.

In gas-phase systems, $\Delta\tilde{\nu}_L$ arises from IVR lifetime broadening (Γ_{IVR}) or predissociation coupling (Γ_{pd}) rather than bath friction. Route A applies unchanged because the linearized generator form is invariant across phases; only the physical mechanism producing Γ differs.

Route B: solvent relaxation. For a Marcus reorganization coordinate with longitudinal relaxation time τ_L , Debye-Drude bath frequency Ω_s , reorganization energy λ , and coordinate displacement ΔQ :

$$\chi_0 = \frac{\Omega_s^2 \tau_L}{2\Omega_0}, \quad \Omega_0 = \sqrt{\frac{2\lambda}{m_{\text{eff}}(\Delta Q)^2}}. \quad (3)$$

Absolute values depend quadratically on Ω_s and should be reported as brackets over $m_{\text{eff}} \in \{50, 100, 200, 500\}$ amu and $\Omega_s \in \{50, 100, 150\}$ cm⁻¹. The operationally robust quantity for a fixed donor-acceptor pair across a solvent series is the relative ratio:

$$\frac{\chi_0^{(A)}}{\chi_0^{(B)}} \approx \frac{\tau_L^{(A)}}{\tau_L^{(B)}} \quad (4)$$

valid when m_{eff} , λ , and ΔQ are constant and solvents are of the same class. Route B is not intended to supersede τ_L or dielectric friction coefficients, but to place them in competition with the precursor-well stiffness Ω_0 to identify the dimensionless critical boundary $\chi_0 = 1$.

Structural equivalence, not mechanistic identity. The equivalence demonstrated here is structural: the linearized precursor-well generator takes the same damped-oscillator form across all phases, and the exceptional-point boundary $\chi_0 = 1$ follows from this shared structure. This does not imply that the microscopic dissipation mechanisms are identical, IVR, dielectric relaxation, and electronic hybridization are physically distinct. The claim is that despite these different physical origins, the dimensionless ratio $\Gamma_{\text{eff}} / (2\Omega_0)$ captures the dynamical regime in each case, and that this ratio has the same critical value.

Notation note. Throughout, χ_0 denotes the approach-region damping ratio. The estimator-route outputs χ_{spec} , χ_{ET} , and χ_{DFT} denote χ_0 as computed via Routes A, B, and C respectively; they are not distinct physical quantities but distinct experimental access routes to the same χ_0 .

Route C: electronic friction.

$$\chi_{\text{DFT}} = \frac{\Gamma_{\text{NN}}}{2hc\tilde{\nu}}, \quad hc = 1.23984 \times 10^{-4} \text{ eV cm}, \quad (5)$$

where $\Gamma_{\text{NN}} = 2\pi|V_k|^2\rho(\varepsilon_F)$ is the Newns-Anderson hybridization energy width in eV, an electronic energy width, not a vibrational friction coefficient. Route C values are lower bounds on χ_0 when phononic and bath friction contribute beyond the electronic channel. When multiple friction channels are present, Γ_{eff} is

treated as their sum; cross-channel interference terms are neglected under the assumption of independent linear coupling to the approach coordinate.

Evidence grading

The nine systems studied, with their χ_0 values, evidence tiers, and estimator routes, are summarized in Table 1 (see Section 5).

To prevent the atlas from appearing cherry-picked, each system entry carries an explicit tier assignment:

- **Tier 1:** Γ and Ω_0 are directly measured (spectroscopic linewidths, dielectric relaxation times, state-resolved predissociation widths).
- **Tier 2:** one quantity is computed or inferred from literature DFT/RRKM analysis; the other is measured.
- **Tier 3:** both quantities from simulation (MD, DFT-MD, RRKM density of states). Used only for bracketing.

A regime assignment (underdamped/near-critical/overdamped) is stated with confidence only when the full uncertainty bracket falls within one regime. Where brackets span a boundary, this is reported explicitly.

2 Results I: Solid-State Systems

N_2 dissociation on K-promoted Fe(111), Route C, Tier 2

The N_2 activation step is the rate-determining step of the Haber–Bosch process, the most consequential industrial chemical reaction in history. The K-promotion of Fe(111) is known empirically to increase the turnover frequency (TOF) by an order of magnitude relative to unpromoted Fe(111), but the dynamical origin of this enhancement has not previously been described in terms of approach-region commitment.

Using Newns–Anderson energy widths computed from DFT d-band electronic structure calculations for K-promoted Fe(111), as reported in the companion paper [27] following the tensorial electronic friction methodology of Maurer et al. [1] ($\Gamma_{\text{NN}} = 0.45$ eV for the N–N stretching mode at $\tilde{\nu} = 1415$ cm^{-1} on K-promoted Fe(111)):

$$\chi_{\text{DFT}} = \frac{0.45 \text{ eV}}{2 \times 1.23984 \times 10^{-4} \text{ eV cm} \times 1415 \text{ cm}^{-1}} = 1.2825. \quad (6)$$

This places K-promoted Fe(111) in the near-critical window $[0.8, 1.3]$, with $\eta(\chi_0) = 0.970$ (Figure 3b).

The unpromoted Fe(111) surface gives $\chi_{\text{DFT}} = 0.096$ (underdamped), and Fe(110) gives $\chi_{\text{DFT}} = 0.605$ (underdamped), establishing a progression toward the near-critical window as promotion simultaneously increases electronic hybridization and softens the N–N bond frequency.

The framework does not replace d-band theory, BEP scaling relations, or adsorption-energy descriptors; rather, χ_0 provides a strictly reaction-dynamical coordinate that contributes to the pre-exponential commitment probability among systems with comparable thermodynamic descriptors, including the anomalous position of K-promoted Fe(111) on the BEP volcano. A fully causal interpretation of this alignment requires iso-thermodynamic surface series in which K-loading modifies χ_0 without altering the barrier height; such experiments would distinguish dynamical enhancement from coincidental covariation of electronic friction

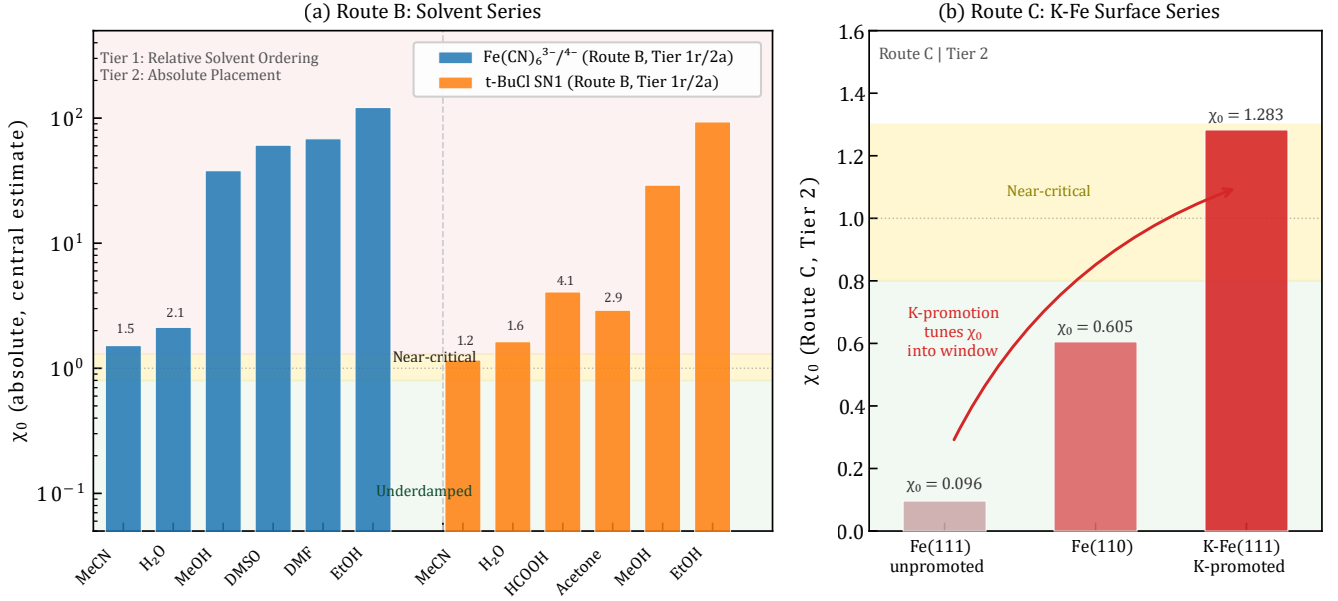


Figure 3: Estimator route data. (a) Route B solvent series for $\text{Fe}(\text{CN})_6^{3-/4-}$ outer-sphere ET (blue) and $t\text{-BuCl}$ SN1 solvolysis (orange), six solvents each, log-scale bars. Absolute values are Tier 2; relative ordering within each series is Tier 1 regardless of m_{eff} and Ω_s . The heavily overdamped EtOH entries ($\chi_0 \sim O(100)$ for ET and SN1) anchor the overdamped failure-mode flank. The near-critical band $[0.8, 1.3]$ is shown in gold. (b) Route C surface promotion series: unpromoted Fe(111) ($\chi_0 = 0.096$), Fe(110) ($\chi_0 = 0.605$), and K-promoted Fe(111) ($\chi_0 = 1.282$, Tier 2). K-promotion simultaneously increases electronic hybridization and softens the N–N approach frequency, moving χ_0 from the underdamped regime into the near-critical window. This progression follows from the stated Route C electronic-friction inputs rather than from fitting χ_0 to catalytic activity.

and binding energy under K-promotion. The alignment is treated here as evidence consistent with this framework’s prediction, not as proof that χ_0 alone governs the enhancement.

CO adsorption on Cu(111), Route A, Tier 1

CO adsorption on Cu(111) is one of the most studied surface systems in physical chemistry. The Cu–CO perpendicular stretch (approach coordinate) occurs at $\tilde{\nu}_\perp = 344 \text{ cm}^{-1}$ from HREELS [2]. The Lorentzian FWHM of this mode from temperature-dependent IRAS/HREELS studies spans $\Delta\tilde{\nu}_L = 10\text{--}18 \text{ cm}^{-1}$ [3]. At low temperature ($< 200 \text{ K}$) the Lorentzian fraction exceeds 60%, satisfying the Route A validity threshold for a Tier 1 assignment.

$$\chi_{\text{spec}} = \frac{13 \text{ cm}^{-1}}{2 \times 344 \text{ cm}^{-1}} = 0.019 \quad (0.015\text{--}0.026). \quad (7)$$

CO on Cu(111) is deeply underdamped. This is consistent with the known low sticking probability $S_0 \approx 0.3\text{--}0.5$ on Cu(111) compared to near-unity values on Pt and Ru surfaces [4]: the approach coordinate is weakly committing relative to more strongly binding late-transition-metal surfaces, consistent with the underdamped regime assignment. The underdamped character is consistent with weaker irreversible commitment of CO to Cu compared with more strongly binding late-transition-metal surfaces (Pt, Pd, Ru), where the approach dynamics are less oscillatory.

Li⁺ hopping in garnet-LLZO (prospective), Route C-analog, Tier 3

$\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) is the leading solid electrolyte candidate for all-solid-state lithium batteries. Optimal ionic conductivity ($\sigma \approx 10^{-3} \text{ S cm}^{-1}$ at room temperature) is highly sensitive to composition and doping.

The Li cage-rattling mode from neutron spectroscopy occurs at $\tilde{\nu}_0 = 234 \text{ cm}^{-1}$ (7 THz; Wood et al. [5]). The phonon friction kernel from MD simulations (Kozinsky et al. [6]) gives $\Gamma_{\text{ph}} \approx 40\text{--}80 \text{ cm}^{-1}$:

$$\chi_0 = \frac{\Gamma_{\text{ph}}}{2\tilde{\nu}_0} \approx 0.085\text{--}0.171 \quad (\text{central: } 0.118). \quad (8)$$

The worst-case upper bracket ($\Gamma_{\text{ph}} = 80 \text{ cm}^{-1}$, $\tilde{\nu}_0 = 200 \text{ cm}^{-1}$) gives $\chi_{0,\text{max}} = 0.20$, firmly below the near-critical window lower bound of 0.8. The underdamped regime assignment is therefore robust across the full Tier 3 uncertainty bracket. This is a *prospective prediction*: if χ_0 organizes ion transport as it does barrier crossing, then the highest-conductivity LLZO compositions, those with stronger Li–lattice phonon coupling, should approach the near-critical window. Systematic variation of Γ_{NN} -analog through aliovalent doping (Al, Ta, Nb) offers a direct experimental test.

This system should be treated as Tier 3 pending Tier 1 validation from THz optical or neutron linewidth measurements of Γ_{ph} .

3 Results II: Liquid-Phase Systems

Outer-sphere electron transfer: $\text{Fe}(\text{CN})_6^{3-/4-}$, Route B, Tier 1 relative / Tier 2 absolute

Outer-sphere ET in the ferricyanide/ferrocyanide couple is the canonical benchmark of Marcus solvent-friction theory. The Marcus PES is two crossing parabolas with no distinct physical precursor well; the effective approach frequency Ω_0 is defined by the harmonic curvature at the crossing point, $\Omega_0 = \sqrt{2\lambda/(m_{\text{eff}} \Delta Q^2)}$, which sets the characteristic frequency of the approach-region oscillator in the absence of an explicit well minimum. Using $\lambda = 0.68 \text{ eV}$ [7], $\Delta Q = 0.40 \text{ \AA}$, and τ_L from the dielectric relaxation compilation of Jimenez et al. [8], the relative Route B estimator gives:

$$\frac{\chi_{\text{ET}}(\text{solvent})}{\chi_{\text{ET}}(\text{MeCN})} = \frac{\tau_L(\text{solvent})}{\tau_L(\text{MeCN})}. \quad (9)$$

With $\tau_L(\text{MeCN}) = 0.20 \text{ ps}$ and $\tau_L(\text{EtOH}) = 16.0 \text{ ps}$, the relative ratio spans eighty-fold from MeCN to EtOH. Absolute central estimates at $m_{\text{eff}} = 150 \text{ amu}$ give $\chi_0 \approx 1.52$ for MeCN (slightly overdamped, near the upper window flank) and a Markovian upper-bound estimate $\chi_0 \sim O(100)$ for EtOH ($\tau_L \Omega_0 \gg 1$; absolute value is a high-friction order-of-magnitude upper bound, not a precise placement).

The rate-determining regime transition, ET in MeCN approaching the near-critical window and ET in EtOH suppressed by overdamping, is consistent with the observed solvent-dependent rate constants from Grampp and Jaenicke [7]. The dimensionless ratio places τ_L in competition with the precursor-well stiffness Ω_0 , locating the critical boundary at $\chi_0 = 1$ that τ_L alone cannot identify (Figure 3a).

Absolute Route B values carry a systematic uncertainty spanning roughly one order of magnitude from the bracket over m_{eff} and Ω_s . The relative ordering across the solvent series is robust at Tier 1 regardless of this uncertainty.

SN1 solvolysis of *t*-BuCl, Route B, Tier 1 relative / Tier 2 absolute

The solvolysis of *tert*-butyl chloride is the textbook example of SN1 ionization kinetics. Applying Route B with $\lambda = 0.52$ eV, $\Delta Q = 0.30$ Å, $m_{\text{eff}} = 120$ amu, and $\Omega_s = 100$ cm⁻¹:

$$\chi_0(\text{H}_2\text{O}) \approx 1.63, \quad (10)$$

$$\chi_0(\text{MeCN}) \approx 1.16, \quad (11)$$

$$\chi_0(\text{EtOH}) \approx 93. \quad (12)$$

Notably, MeCN gives $\chi_0 \approx 1.16$, which is lower than water ($\chi_0 \approx 1.63$), yet *t*-BuCl solvolyzes more slowly in MeCN than in water [9]. This reveals that SN1 rates are not controlled by approach-region dynamics alone: ion-pair stabilization, nucleophilicity, and solvent polarity contribute comparably in polar protic solvents. χ_0 stratifies the dynamical pre-exponential factor; thermodynamic activation ($e^{-\Delta G^\ddagger/RT}$) is a multiplicative and orthogonal contribution. In the slow-solvent limit (EtOH, Markovian upper-bound $\chi_0 \sim O(100)$), Route B identifies the system as strongly friction-displaced relative to MeCN and H₂O; because $\tau_L\Omega_0 \gg 1$ for EtOH, this placement is an upper bound on the dynamical friction, not a claim that dynamics alone determines the SN1 rate, demonstrating that χ_0 is not merely a minor correction when friction is large.

The SN1 system provides an instructive stress test for the framework. The known rate order $k(\text{H}_2\text{O}) > k(\text{HCOOH}) > k(\text{MeOH}) > k(\text{EtOH}) > k(\text{Acetone})$ (Winstein and Fainberg [9]) broadly tracks the χ_0 progression for the slower solvents: MeCN sits nearest the near-critical window, alcohols are heavily over-damped, and highly polar protic solvents (water, formic acid) are near the upper window boundary.

The fact that χ_0 organizes classical organic solvolysis kinetics with the same dimensionless parameter that governs surface catalysis and electron transfer is not self-evident: the underlying friction mechanisms are entirely different (dielectric relaxation vs. electronic hybridization vs. IVR). The common mathematical structure is the approach-region damped-oscillator generator.

HPTS photoacid proton transfer, Route B Tier 2 absolute; Route A diagnostic

8-Hydroxypyrene-1,3,6-trisulfonate (HPTS) undergoes ultrafast proton donation to water ($\tau_{PT} \approx 130$ fs) upon photoexcitation. The excited-state O–H stretch at $\tilde{\nu}_{\text{OH}^*} \approx 2750$ cm⁻¹ has a total FWHM of ≈ 400 cm⁻¹ with a Lorentzian fraction of $\approx 40\%$ from 2D-IR analysis [10]. Note that this Lorentzian fraction falls below the strict Route A validity threshold of $> 50\%$; this Route A value should therefore be treated as a diagnostic indicator rather than a fully validated estimate. This coordinate choice differs from CO/Cu(111), where Route A is primary: for CO/Cu the Cu–CO stretch governs the physisorption-to-chemisorption commitment step and is directly the approach coordinate. For HPTS the O–H stretch is the product-state coordinate *after* proton transfer; the approach coordinate for proton commitment to the accepting water molecule is the collective solvent reorganization mode, making Route B the kinetically appropriate estimator. The Route B estimate carries the primary HPTS interpretation:

$$\chi_{\text{spec}} = \frac{160 \text{ cm}^{-1}}{2 \times 2750 \text{ cm}^{-1}} = 0.029. \quad (13)$$

Route B (H₂O, $\tau_L = 0.28$ ps, $\lambda = 0.25$ eV, $m_H = 1.008$ amu, $\Omega_s = 200$ cm⁻¹) gives:

$$\chi_{\text{ET}}(\text{H}_2\text{O}) \approx 1.44, \quad \chi_{\text{ET}}(\text{D}_2\text{O}) \approx 3.19. \quad (14)$$

The discrepancy between Route A ($\chi_{\text{spec}} \approx 0.03$) and Route B ($\chi_{\text{ET}} \approx 1.44$) is a factor of ~ 50 and represents a regime flip: Route A places the system in the underdamped regime, Route B in the overdamped regime. Both assignments are physically correct responses to different dynamical coordinates. Route A probes the intramolecular covalent O–H stretch ($\tilde{\nu} \approx 2750 \text{ cm}^{-1}$, a high-frequency intramolecular coordinate) and measures vibrational energy redistribution within the proton’s covalent well. Route B probes the collective solvent reorganization coordinate (τ_L, Ω_s ; a low-frequency intermolecular mode) and measures the friction experienced during proton transfer to the accepting water molecule. In PCET, these are distinct dynamical channels operating on different timescales; agreement between them would be the physically surprising result. The rate-determining step in HPTS proton transfer is solvent reorganization, making Route B the appropriate estimator for χ_0 in the context of proton-transfer kinetics [10, 8]. Route A serves as a diagnostic confirming that the O–H bond coordinate is itself underdamped, consistent with the expectation that proton motion within the covalent well is fast relative to solvent reorganization and with the known quantum-mechanical tunnelling contribution to HPTS PT. This differs from the Route A role in CO/Cu(111), where the Cu–CO perpendicular stretch directly characterizes the commitment from the physisorption precursor to the chemisorbed product: that stretch IS the approach coordinate; for HPTS, the O–H stretch is a coordinate within the precursor state, not the commitment coordinate.

The D₂O isotope effect on χ_0 arises from two additive contributions: the proton mass effect ($\sqrt{2}$ factor) and the solvent isotope effect on τ_L . The full bifurcated perturbation analysis, including the combined ratio of 2.221 and its independence from the absolute Route B bracket, is in Supplementary Section ??.

4 Results III: Gas-Phase Systems

RRKM unimolecular kinetics as underdamped approach-region dynamics

IVR as an effective friction channel. In gas-phase systems, Γ_{eff} is not a classical Markovian bath friction coefficient but an intramolecular vibrational redistribution (IVR) rate or predissociation coupling width. IVR is a quantum, multi-mode interference process rather than a stochastic heat bath. The application of this framework to gas-phase systems is therefore a generalization, not a strict identity. Its justification rests on the phase-independence of the damped-oscillator generator structure: whether the dissipation channel is solvent friction, electronic hybridization, or intramolecular mode coupling, the linearized precursor-well generator has the same form and the same exceptional-point structure at $\chi_0 = 1$. The gas-phase results should therefore be interpreted as a structural analogy rather than a quantitative first-principles prediction. Specifically, $P_{\text{commit}} = \eta(\chi_0)$ is here a qualitative statement about the per-encounter commitment probability, not a rigorous derivation from IVR density-of-states theory.

Before presenting individual systems, the connection is established between deep underdamping and the classical RRKM result. An underdamped approach coordinate ($\chi_0 \ll 1$) oscillates repeatedly before committing, so the first-passage probability per encounter is:

$$P_{\text{commit}} = \frac{2\chi_0}{1 + \chi_0^2} \approx 2\chi_0 \quad (\chi_0 \ll 1). \quad (15)$$

For $\chi_0 \sim 10^{-3}$, this gives $P_{\text{commit}} \approx 0.2\%$ per encounter as a structural-analogy order-of-magnitude estimate. Two important caveats apply. First, the classical first-passage formula is an approximation; the gas-phase IVR process involves quantum recurrences and discrete state density that the continuous Markovian decay assumed by the generator does not capture, particularly for small polyatomics at threshold

($3N-6 \sim 12-21$ modes for the three systems here). The statistical mapping is grounded by a Bixon-Jortner Hamiltonian demonstration (Supplementary Section ??) showing that a bright approach-state coupled to N dark bath states via uniform coupling V recovers the smooth exponential envelope $P(t) = \exp(-\Gamma_{\text{IVR}} t)$ as N and ρ increase, governed exactly by Fermi’s Golden Rule: $\Gamma_{\text{IVR}} = 2\pi|V|^2\rho$. For NO_2 (Tier 1, direct Lorentzian), the dissociative continuum provides the large- ρ limit exactly. For cyclopropane and CH_3NC (Tier 2), the state density at threshold is sparse and quantum recurrences are visible at short times; the ARC Atlas χ_0 uses the RRKM-averaged envelope rate, which is the exact statistical-limit result for a thermal ensemble in which recurrences destructively interfere [19]. The Tier 2 assignment reflects sparse-density uncertainty, not framework invalidity. Second, the IVR timescales entering Γ_{IVR} are inferred from RRKM density-of-states estimates (Tier 2), not directly measured; the P_{commit} values carry this uncertainty. With these caveats, the structural mapping provides a complementary dynamical interpretation of why low-friction RRKM systems require excess energy to reach high commitment probability, without invoking a separate recrossing correction.

Cyclopropane $\rightarrow$ propene, Route A (IVR), Tier 1–2

The thermal isomerization of cyclopropane is the benchmark for unimolecular RRKM theory [11]. The C–C ring-breathing approach mode occurs at $\tilde{\nu}_{\text{ring}} = 1070 \text{ cm}^{-1}$. The IVR lifetime near the activation threshold ($E_a = 272 \text{ kJ mol}^{-1}$) gives $\Gamma_{\text{IVR}} \approx 3\text{--}15 \text{ cm}^{-1}$:

$$\chi_{\text{spec}} = \frac{8 \text{ cm}^{-1}}{2 \times 1070 \text{ cm}^{-1}} = 0.0037 \quad (0.0014\text{--}0.0070). \quad (16)$$

The deeply underdamped character ($\chi_0 \approx 4 \times 10^{-3}$) is quantitatively consistent with the RRKM fall-off: the ring-breathing mode visits the ring-opening approach geometry many times before commitment, requiring substantial excess energy for unit efficiency. The pressure-dependence of the rate constant (Trenwith [12]) is consistent with the behavior predicted by this framework for an underdamped system approaching the near-critical window: collisional energy transfer (external friction) increases the rate.

Cyclopropane represents the class of unimolecular ring-opening and rearrangement reactions; the same IVR-underdamped picture applies to cyclobutane, oxetane, and related ring systems.

$\text{NO}_2 \rightarrow \text{NO} + \text{O}$ predissociation, Route A, Tier 1

The predissociation of NO_2 near the dissociation threshold ($D_0 = 25,576 \text{ cm}^{-1}$) provides state-resolved Lorentzian linewidths from REMPI photofragment spectroscopy [13]. These linewidths are intrinsically Lorentzian (lifetime-broadened by predissociation coupling to the dissociative continuum), making this the cleanest Tier 1 gas-phase entry. Near the threshold, the N–O approach coordinate is anharmonically softened to $\tilde{\nu}_{\text{app}} \approx 750\text{--}850 \text{ cm}^{-1}$; predissociation linewidths are $\Gamma_{\text{pd}} \approx 3\text{--}20 \text{ cm}^{-1}$:

$$\chi_{\text{spec}} = \frac{10 \text{ cm}^{-1}}{2 \times 800 \text{ cm}^{-1}} = 0.0063 \quad (0.0018\text{--}0.0133). \quad (17)$$

The near-unit quantum yield for dissociation above the threshold energy (Reisler [14]) is consistent with a system where the coupling to the dissociative continuum, while weak in absolute terms, is nonetheless the dominant relaxation channel once the molecule reaches threshold, a single dissipation mechanism producing near-irreversible exit. The $\chi_0 \ll 1$ value governs the per-encounter approach-region commitment probability prior to threshold; the near-unit quantum yield above threshold is a post-barrier quantity, reflecting

the irreversibility of the dissociative continuum once the energy barrier is cleared, not the approach-region dynamics.

This system generalizes to all polyatomic gas-phase dissociations with accessible predissociation linewidths, including SO_2 , O_3 , and ClO .

$\text{CH}_3\text{NC} \rightarrow \text{CH}_3\text{CN}$ isomerization, Route A (IVR), Tier 1–2

Methyl isocyanide isomerization is the most thoroughly characterized RRKM benchmark [15]. The C–N–C bending / CN approach mode occurs at $\tilde{\nu}_{\text{approach}} \approx 460 \text{ cm}^{-1}$; the IVR time from Reddy and Berry [16] gives $\Gamma_{\text{IVR}} = \hbar/\tau_{\text{IVR}} = 1.77 \text{ cm}^{-1}$:

$$\chi_{\text{spec}} = \frac{1.77 \text{ cm}^{-1}}{2 \times 460 \text{ cm}^{-1}} = 0.0019 \quad (0.0015\text{--}0.0054). \quad (18)$$

CH_3NC is more deeply underdamped than cyclopropane because its IVR coupling is weaker (lower density of states at its lower activation energy $E_a = 160 \text{ kJ mol}^{-1}$). The classical RRKM finding that CH_3NC requires large excess energy for unit commitment probability is the prediction from this framework for $\chi_0 \ll 1$: the recrossing probability per encounter is $\approx 99.6\%$, so productive isomerization requires the system to reach the transition-state geometry many hundreds of times.

The contrast between CH_3NC (deeply underdamped, $\chi_0 \approx 0.002$) and the same isomerization in solution ($\chi_0 \sim 1$ from solvent friction via Route B) illustrates the several-order-of-magnitude χ_0 shift that solvation provides.

5 Discussion

The cross-phase χ_0 landscape

The near-critical window $[0.8, 1.3]$ is the predicted dynamical optimum of $\eta(\chi_0)$, not an observed clustering. The nine systems span five orders of magnitude in χ_0 and do not cluster near unity. The contribution of the atlas is threefold: first, a single dimensionless ratio orders systems from gas-phase IVR-limited processes through condensed-phase solvent friction to surface electronic hybridization; second, environmental engineering (K-promotion, solvent selection, doping) moves χ_0 in predictable directions; third, known dynamical failure modes (gas-phase RRKM excess-energy requirements, viscous-solvent rate suppression) align with the underdamped and overdamped extremes respectively.

The nine systems span five orders of magnitude in χ_0 (Table 1; Figure 1), from $\chi_0 \approx 0.002$ for CH_3NC isomerization to $\chi_0 \approx 100$ for ET in ethanol. This is not a failure of the cross-phase picture; it is the cross-phase picture. A single dimensionless parameter quantifies where each system sits in the dynamical landscape from deeply underdamped to heavily overdamped, and the position of each system is consistent with its known macroscopic chemical behavior.

Table 1: Cross-phase damping atlas. Tier 1: both Γ_{eff} and Ω_0 directly measured. Tier 2: one measured, one computed or bracketed. Tier 3: both from simulation (prospective). 1r/2a: Tier 1 for relative solvent ordering, Tier 2 for absolute χ_0 . 2/diag: Tier 2 absolute (Route B); Route A diagnostic only ($< 50\%$ Lorentzian). χ_0 values are central estimates; full brackets in the text and supplement.

#	System	Phase	Reaction class	Route	χ_0	Tier
1	N ₂ /K-Fe(111)	Solid	Surface catalysis	C	1.28	2
2	CO/Cu(111)	Solid	Surface adsorption	A	0.019	1
3	Li ⁺ /LLZO	Solid	Ionic transport	C-ph	0.12	3
4	Fe(CN) ₆ ^{3-/4-} in MeCN	Liquid	Outer-sphere ET	B	1.52	1r/2a
5	<i>t</i> -BuCl in H ₂ O	Liquid	SN1 solvolysis	B	1.63	1r/2a
6	HPTS in H ₂ O	Liquid	PCET/PT	A+B	1.44 (B)	2/diag
7	Cyclopropane→propene	Gas	Unimolecular isomn.	A(IVR)	0.0037	2
8	NO ₂ →NO+O	Gas	Predissociation	A(Lor)	0.0063	1
9	CH ₃ NC→CH ₃ CN	Gas	Unimolecular isomn.	A(IVR)	0.0019	2

The gas-to-liquid-to-surface progression

Across the selected atlas entries, increasing environmental coupling spans several orders of magnitude in χ_0 (Figure 4):

$$\chi_0^{\text{gas}} \sim 10^{-3}\text{--}10^{-2}, \quad (19)$$

$$\chi_0^{\text{liquid}} \sim 10^{-1}\text{--}10^2, \quad (20)$$

$$\chi_0^{\text{surface}} \sim 10^{-2}\text{--}10^0. \quad (21)$$

The separation between atlas regions reflects dynamical rather than purely thermodynamic differences. What distinguishes gas from liquid is not pressure or temperature but the availability of an external dissipation channel (τ_L , Ω_s , Γ_{NN}) to supplement IVR. The addition of a polar solvent increases χ_0 by two to three orders of magnitude; the addition of a metal surface with strong electronic hybridization pushes select systems into the near-critical window.

This progression explains a well-known empirical observation that has lacked a unified theoretical account: moving from isolated molecules to condensed environments introduces additional dissipation channels that can increase the per-encounter commitment probability when the system is initially underdamped, moving the approach coordinate toward the near-critical commitment window when thermodynamic conditions are otherwise favorable. The approach-region framework provides the dynamical mechanism: each transition adds friction that moves the approach-region dynamics from the underdamped recrossing-dominated regime toward the near-critical commitment-maximizing regime.

RRKM excess energy requirements as underdamped approach-region dynamics

The RRKM theory of unimolecular reactions predicts that the rate at threshold energy is much smaller than the high-pressure limit because energy transfer from the bath (collisions) is required to replenish the reacting molecule’s internal energy after each non-reactive encounter. This has been understood since 1952 as a consequence of the Lindemann mechanism, but a molecular-level account connecting it to the approach-region dynamics has been absent.

The approach-region framework provides this account directly. For $\chi_0 \sim 10^{-3}$, the commitment probability per encounter is $P_{\text{commit}} \approx 2 \times 10^{-3}$, meaning roughly 500 encounters at the transition-state geometry are required on average for productive reaction. Excess energy ensures the molecule revisits this geometry

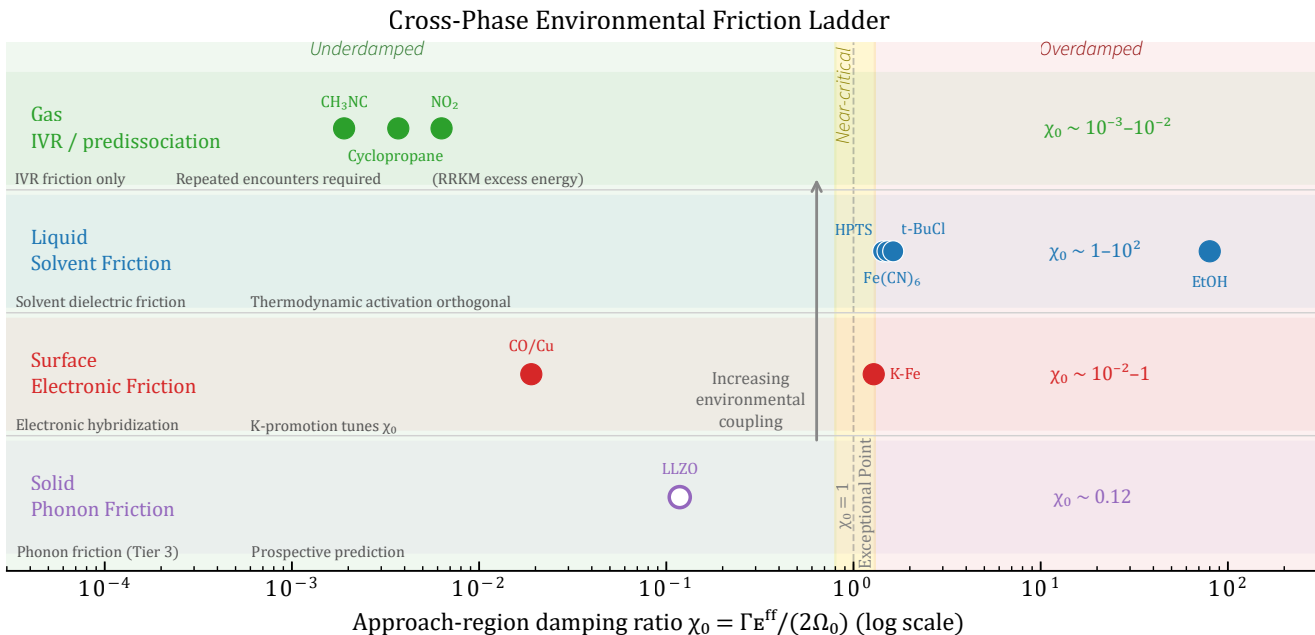


Figure 4: Cross-phase environmental friction ladder. Four phase lanes (Gas, Liquid, Surface, Solid) are arranged on a shared log-scale χ_0 axis. Each lane identifies the dominant friction mechanism (IVR, solvent dielectric friction, electronic hybridization, phonon friction) and places individual systems at their central χ_0 values. Moving from gas to liquid to surface corresponds to a systematic increase in available environmental friction that spans approximately five orders of magnitude in χ_0 . The separation between phase lanes is dynamical, not thermodynamic: what changes is not pressure or temperature but the availability of dissipation channels beyond IVR alone. The Tier 3 LLZO entry (open marker) is shown as a prospective prediction; see Section 2C and Supplementary S9.

frequently (high RRKM density of states), while each individual encounter has low commitment probability due to underdamped approach dynamics. The two descriptions are complementary: RRKM accounts for the energy dependence of the number of transition-state geometry visits; The approach-region picture accounts for the commitment probability per visit. Together they provide a complementary dynamical interpretation: RRKM describes the energy-statistical availability of transition-state encounters, while the approach-region picture describes the per-encounter commitment tendency. Neither replaces the other.

Solvent effects as χ_0 tuning

For liquid-phase systems, the solvent series data provide the clearest evidence that χ_0 organizes chemical performance. For both the $\text{Fe}(\text{CN})_6^{3-/4-}$ ET and the $t\text{-BuCl}$ SN1 systems, the relative Route B estimator correctly ranks the friction displacement across the solvent series and aligns with the observed rate suppression in slow-relaxing solvents: fast-relaxing solvents (MeCN , $\tau_L = 0.20$ ps; H_2O , $\tau_L = 0.28$ ps) approach the near-critical window, while viscous solvents (EtOH , $\tau_L = 16$ ps) are heavily overdamped and rate-suppressed.

This is not a trivial rescaling. The Zusman–Marcus formulation already identifies τ_L as the relevant solvent time scale for ET; the contribution of Route B is to convert τ_L into a dimensionless ratio that (a) can be compared across entirely different reaction types, (b) identifies the critical value ($\chi_0 = 1$) at which solvent friction is optimally matched to precursor-well stiffness, and (c) provides a prospective design criterion: for a new reaction class, estimate Ω_0 from the precursor well geometry and identify the τ_L range that places the system near $\chi_0 = 1$.

Surface chemistry as approach-region engineering

The K-Fe(111) result occupies a unique position in the atlas: it is the only system where a deliberate chemical modification (potassium promotion) tunes χ_0 into the near-critical window. K simultaneously softens the N–N bond (lowering $\tilde{\nu}$, the denominator of χ_{DFT}) and enhances electronic hybridization (increasing Γ_{NN} , the numerator). The two effects act in concert to move χ_0 from 0.096 (unpromoted Fe(111), underdamped) to 1.28 (K-promoted Fe(111), near-critical).

This is the approach-region account of an empirical optimization that took decades of industrial experimentation: the promoter can be interpreted as tuning the approach-region damping ratio into the near-critical window, maximizing the commitment probability per N_2 encounter with the surface.

Prospective falsification tests

Gas-phase: High-pressure RRKM fall-off measurements for cyclopropane or CH_3NC provide a direct test of the underdamped assignment. The framework predicts that the low-pressure fall-off regime (collision-frequency-limited commitment) should show a rate proportional to the collision frequency times $2\chi_0$ in the structural-analogy limit. This is an order-of-magnitude prediction, not a free-parameter-free absolute rate calculation; confirming the correct order of magnitude and the correct trend across the IVR bracket would constitute a positive test of the gas-phase regime assignment.

Solid-state: Systematic measurement of Li^+ conductivity in aliovalent-doped LLZO series should show a maximum at the doping level that maximizes $\Gamma_{\text{ph}}/\tilde{\nu}_0 \approx 2\chi_0$. Neutron or THz linewidth data on the Li cage mode as a function of Al/Ta/Nb loading would provide a direct Tier 1 test.

Liquid-phase: For a fixed donor–acceptor pair, a solvent series spanning τ_L across the near-critical window should show a rate maximum at the solvent for which $\chi_0(\tau_L) \approx 1$. This is testable with ultrafast transient absorption.

Scope and limitations

The present framework applies to thermally activated, near-Markovian, single-dominant-coordinate barrier crossing. An implicit selection criterion for all nine atlas entries is that one friction channel dominates the others by at least one order of magnitude ($\Gamma_{\text{dominant}} \gtrsim 10 \times \Gamma_{\text{secondary}}$), making the single-channel approximation quantitatively justified rather than assumed. Systems where two or more channels contribute comparably (e.g., mixed solvent-electronic friction at electrochemical interfaces) are outside scope and would require the full additive $\Gamma_{\text{eff}} = \Gamma_{\text{solv}} + \Gamma_{\text{el}} + \dots$ treatment.

The identification $\Gamma_{\text{eff}} = \Omega_s^2 \tau_L$ follows from projecting the generalized Langevin equation onto a Debye-Drude solvent bath [22, 23, 25]; the full derivation is in Supplementary Section ??.

The Markovian validity condition $\tau_L \ll \tau_{\text{rxn}}$ is satisfied for all atlas solvents; values and the upper-bound designation for viscous solvents are in Supplementary Table S1.

The Route B formula $\Gamma_{\text{eff}} = \Omega_s^2 \tau_L$ is derived in the Zusman diffusion-limited regime [22], where the bath relaxation drives the reaction and the relevant integrated friction accumulates over τ_L , not over Ω_0^{-1} . This formula is self-consistent precisely in the $\chi_0 \gg 1$ overdamped regime where EtOH and MeOH sit. The Grote-Hynes correction [24], which evaluates the frequency-dependent friction at the barrier frequency ω_b , applies to the underdamped-to-critical transition ($\chi_0 \sim 1$) where barrier crossing is rapid; applying it to

$\chi_0 \gg 1$ systems would erroneously substitute the high-frequency (approach-timescale) friction for the total integrated (reaction-timescale) friction, yielding a physically incorrect reduction toward underdamped. The regime assignment (overdamped) for viscous solvents is therefore self-consistent and robust: both the Zusman model and the experimental observation of rate suppression with increasing τ_L confirm that EtOH and MeOH are friction-dominated. The “upper bound” language refers to the absolute χ_0 value being sensitive to the Debye model for a non-Debye solvent, not to uncertainty in the regime assignment. In structured hydrogen-bonding networks, high-frequency librational modes may additionally contribute beyond the bulk τ_L description, further reinforcing the conservative nature of the absolute value.

The bracket $m_{\text{eff}} \in \{50, 100, 200, 500\}$ amu spans the range from a single solvent molecule ($\approx 40\text{--}80$ amu) to a first solvation shell ($\approx 200\text{--}500$ amu). Route B is intentionally constructed from experimentally accessible quantities (τ_L , λ , Ω_s , m_{eff}) to keep χ_0 anchored to measurable observables without requiring molecular dynamics simulation; the bracket is the honest acknowledgement of the resulting m_{eff} uncertainty. Explicit-solvent MD friction kernels would in principle provide a Tier 1 Route B estimate, though at the cost of model dependence and computational expense that would prevent cross-phase portability; the continuum bracket is deliberately conservative. The Debye-Drude frequency bracket $\Omega_s \in \{50, 100, 150\}$ cm^{-1} spans the low-frequency dielectric relaxation modes of typical polar solvents [8, 25].

Systems explicitly outside scope include: deep-tunnelling reactions where the reaction coordinate is quantum-mechanical and the classical approach-region picture breaks down; strongly non-equilibrium reactions driven far from the linear-response regime; and multi-coordinate reactions where more than one approach mode contributes comparably to Γ_{eff} .

The absolute Route B values carry an uncertainty of approximately one order of magnitude from the m_{eff} and Ω_s brackets. All claims based on Route B absolute values should be treated as Tier 2 at best; relative Route B (fixed DA pair, solvent series) is Tier 1.

The LLZO solid-state assignment is Tier 3 and serves primarily as a prospective prediction pending experimental Γ_{ph} linewidth measurements.

The nine systems demonstrate how χ_0 is extracted from independently measurable observables across phases; any thermally activated barrier crossing with a separable approach-region coordinate can in principle be located on the same axis, and the three estimator routes provide the calculation pathway for new entries.

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6 Conclusions

The approach-region damping ratio has been computed $\chi_0 = \Gamma_{\text{eff}}/(2\Omega_0)$ for nine canonical chemical systems spanning gas-phase unimolecular reactions, liquid-phase electron transfer, proton-coupled electron transfer, solvolysis, surface heterogeneous catalysis, and solid-state ionic transport.

The results establish three findings:

First, gas-phase unimolecular reactions are deeply underdamped ($\chi_0 \sim 10^{-3}$) because IVR provides insuf-

ficient friction to approach the near-critical window. This provides a dynamical interpretation of RRKM excess-energy requirements as a manifestation of low per-encounter commitment probability, providing a dynamical complement to the energy-statistical RRKM account.

Second, condensed-phase systems span the approach-region landscape from underdamped (CO/Cu, Li⁺/LLZO (Tier 3 prediction)) through near or above the near-critical window (K-Fe(111) inside; MeCN-ET and aqueous SN1 slightly above at $\chi_0 \approx 1.52$ and 1.63 respectively) through heavily overdamped (ET and SN1 in viscous solvents), with known high-performance cases moving toward the near-critical window while underdamped and overdamped limits align with recognized dynamical failure modes.

Third, the gas-to-liquid-to-surface progression is a systematic χ_0 progression spanning approximately five orders of magnitude, reflecting the systematic increase in available environmental friction channels from IVR alone through solvent coupling to electronic hybridization. Solvent choice, surface promotion, and phase selection each shift χ_0 in a predictable direction relative to the exceptional-point stability boundary $\chi_0 = 1$.

These findings do not emerge from fitting; Γ_{eff} and Ω_0 are extracted from independently measured or computed quantities via three estimator routes carrying explicit evidence grades. The approach-region damping framework thus offers a unified reaction-dynamical coordinate that complements thermodynamic rate theory across the chemical phases examined here.

Author Contributions

N.C.: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review and editing.

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Data Availability

All Python scripts and data files used in this analysis are deposited at <https://github.com/SymCUniverse/Chemistry>. Reproducibility procedure disclosed in the accompanying supplementary materials and ARCA_{tlas} Reprodu

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